

University of Aizu

Research Paper Writing Seminar I

Where do research ideas come from?

Subramaniam Ganesh

Lecture goals

- Demystify the origin of research ideas
- Inspire curiosity, critical thinking, and methodological precision
- Show how these elements drive scientific advancement

Lecture outline

- Primary sources of research ideas
- Graduate student progression
- Research orientation
- Scientific rigor
- Researcher mindsets
- An example from my research project

Where do research ideas come from?

- How does this work?
- Says who?
- I don't believe that!
- Why are we doing this?
- Why are they both right?
- Can we learn from the abnormal?



Where do research ideas come from?

Curiosity and
Theoretical Exploration:

Newton's Laws of
Motion

Millions saw the apple fall



But only Newton asked why

Where do research ideas come from?

Nature-Inspired
Innovation:

The Japanese Bullet
Train (Shinkansen)

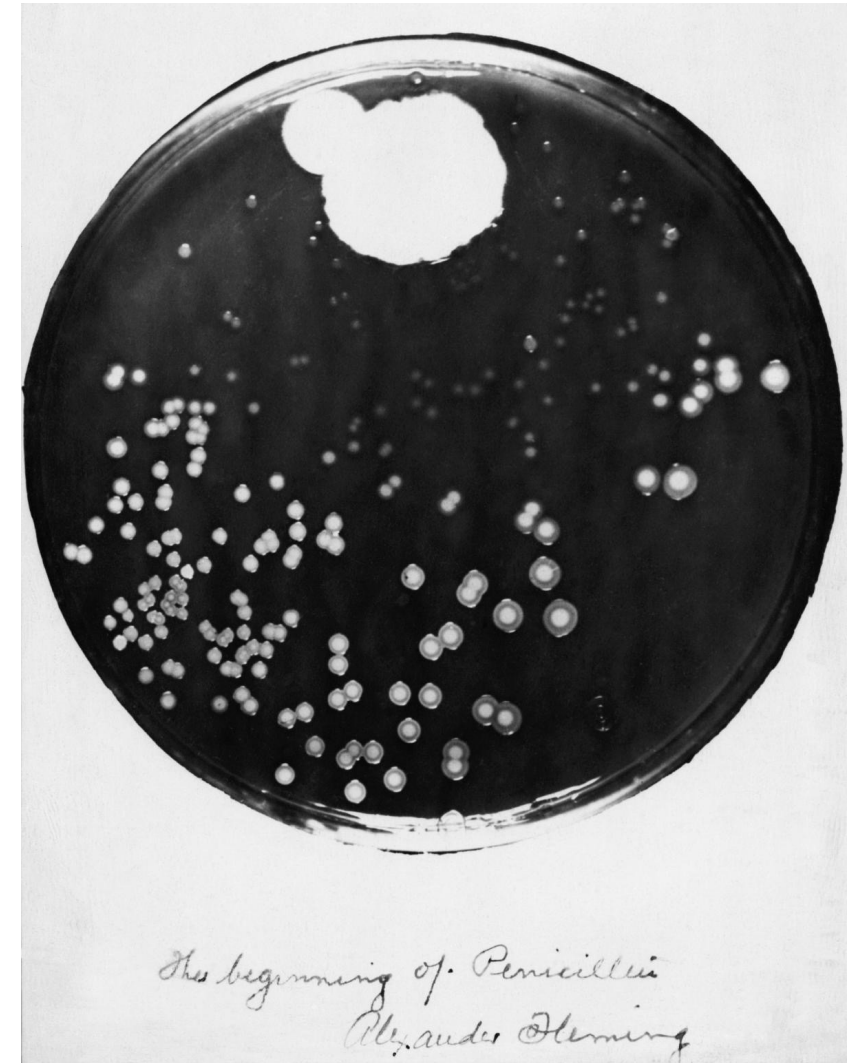
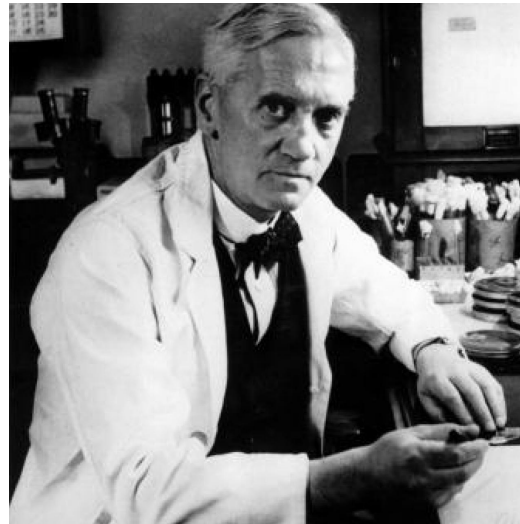


Where do research ideas come from?

Accidental Discovery:

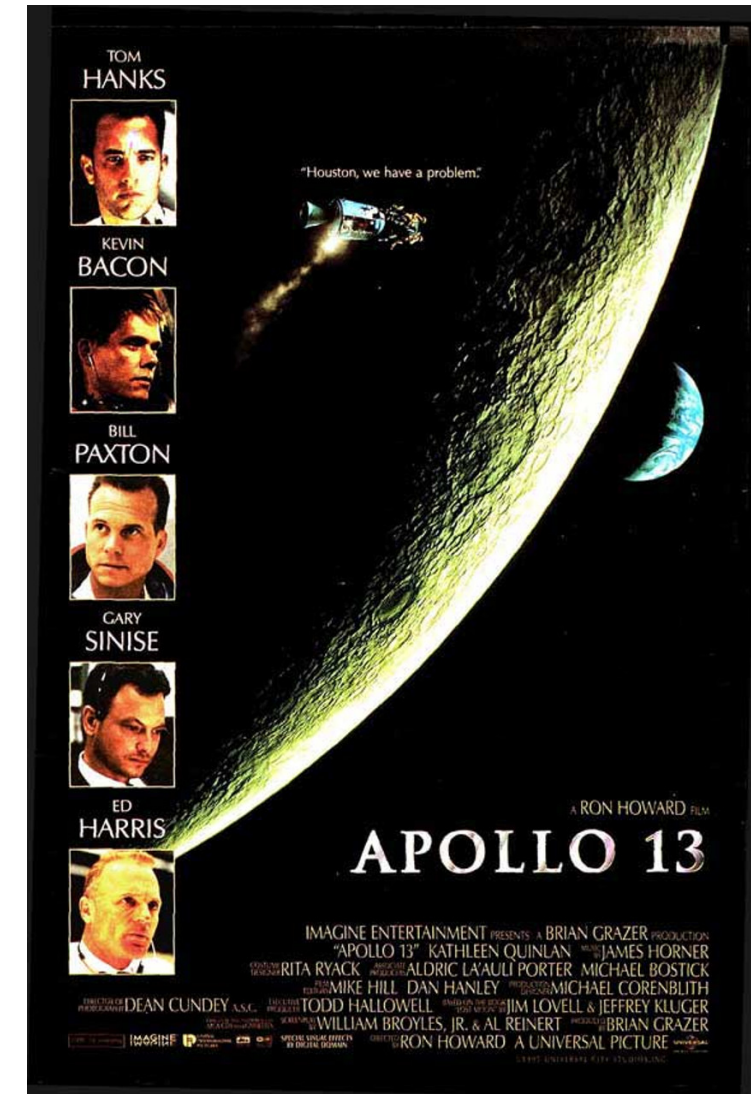
Penicillin

Alexander Fleming



Where do research ideas come from?

Necessity is the Mother of Invention!



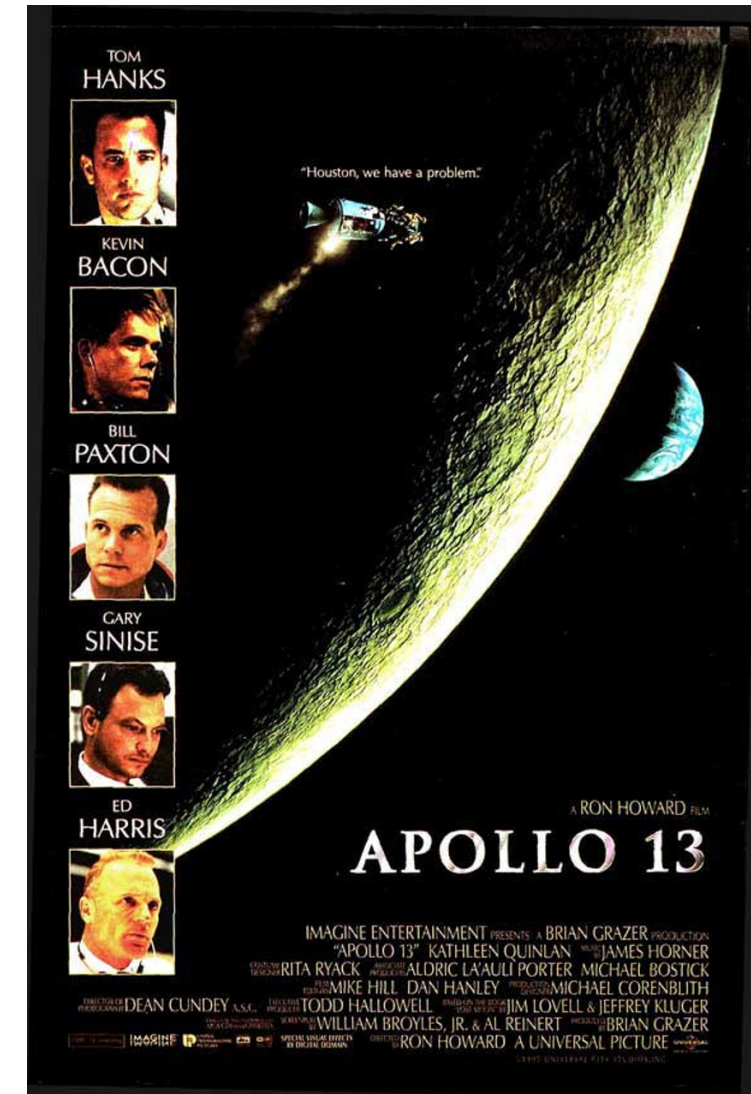
Where do research ideas come from?

Necessity is the Mother of Invention!

An oxygen tank explosion >>

Loss of power, oxygen, and navigation systems made a Moon landing impossible and jeopardized safe return.

Successful Failure!



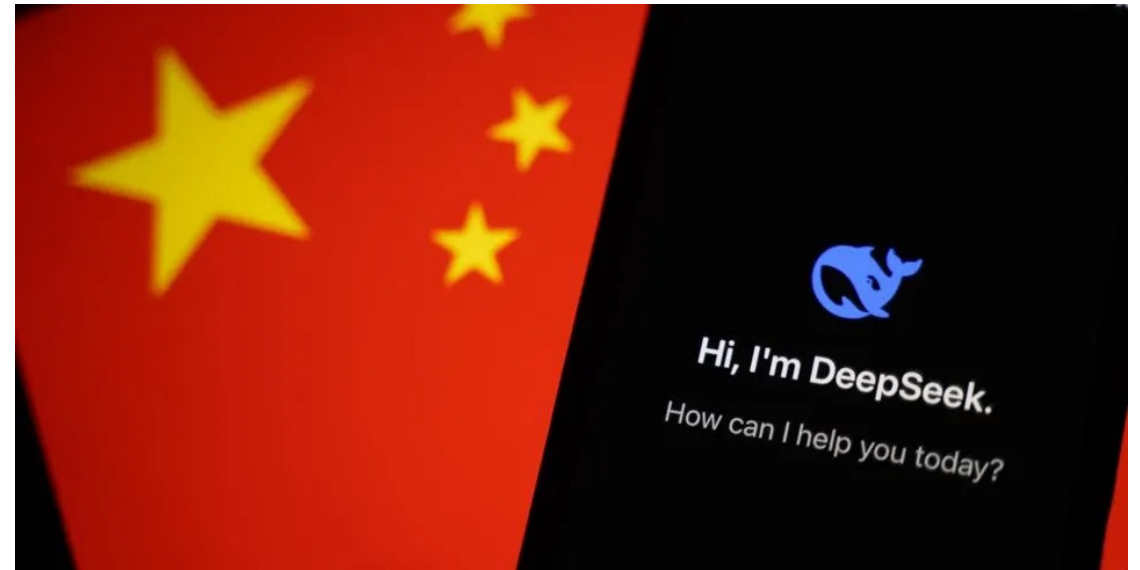
Where do research ideas come from?

Aspiration: National prestige can inspire bold initiatives—at times achieved at a fraction of the cost, compared to other countries, through frugal innovations!

Chandrayaan-3: India's Lunar Leap!

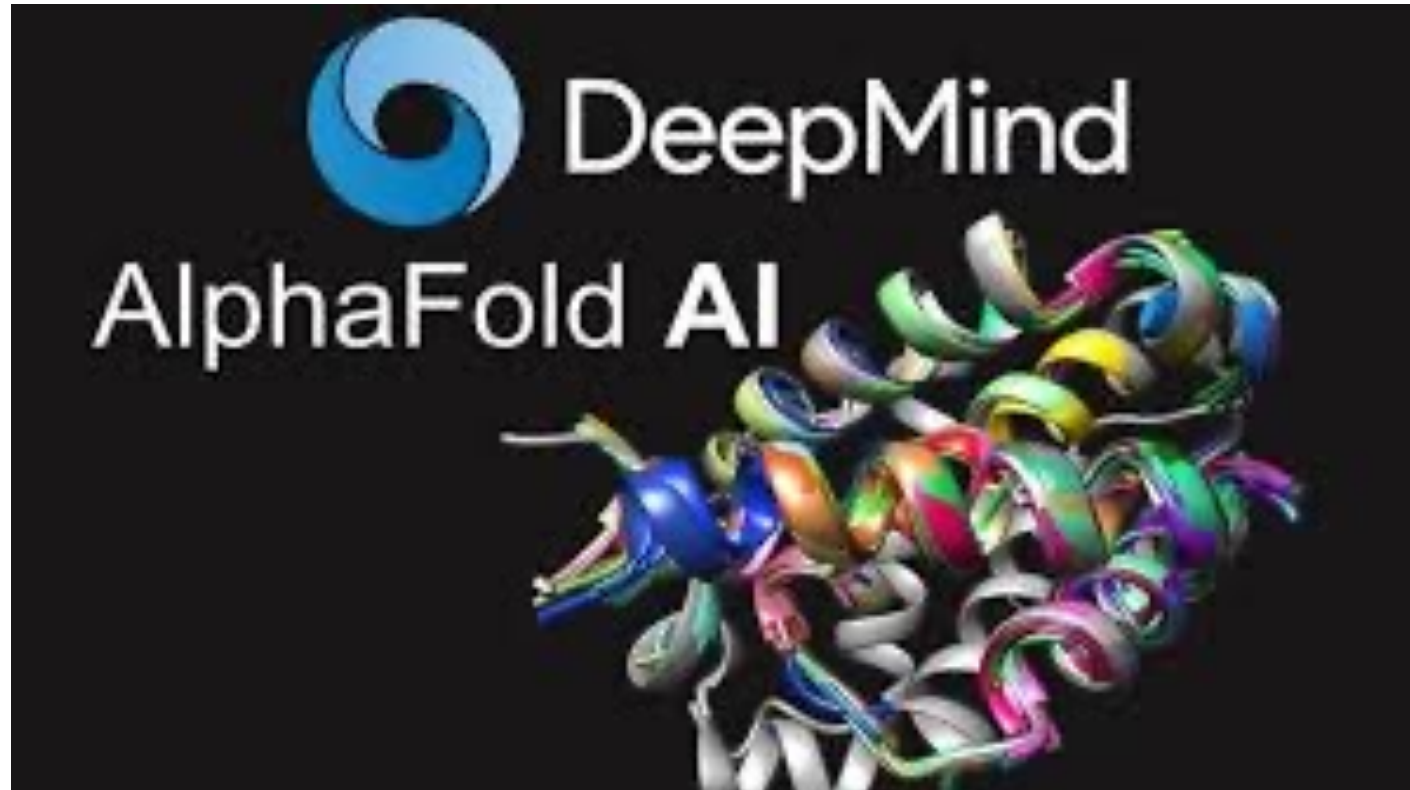


DeepSeek: Chian's AI platform



Where do research ideas come from?

DeepMind's AlphaFold: Biology + Artificial Intelligence



Where do research ideas come from?

Examples showed that research ideas can come from!

- Nature (biomimicry)
- Curiosity and observation
- Accidents and anomalies
- Societal or technological needs
- Cultural or national aspirations
- Cross-disciplinary insights

Where do research ideas come from?

I have six honest serving men
They taught me all I knew
I call them What and Where and When
And How and Why and Who

What?

Where?

When?

Who?

Why?

How?

Where do research ideas come from?

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What?

Where?

When?

Who?

Why?

How?

The sources of ideas can be divided into two:

- Replicating/extending the work of others
- Doing something original

Where do research ideas come from?

Replicating/extending the work of others:

These ideas build on existing studies, theories, or technologies—either by repeating them in new contexts or by adding new variables or methods.

- Replication
- Extension
- Technology Adaptation

Where do research ideas come from?

Replicating/extending the work of others:

These ideas build on existing studies, theories, or technologies—either by repeating them in new contexts or by adding new variables or methods.

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Replicate a cybersecurity experiment that tested phishing email detection rates, this time using a different demographic

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A researcher extends a U.S. study on fitness apps by adding biometric feedback to examine user motivation in Southeast Asia

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- Extension
- Technology Adaptation

Blockchain, initially developed for cryptocurrency, is adapted by governments to create tamper-proof land ownership records.

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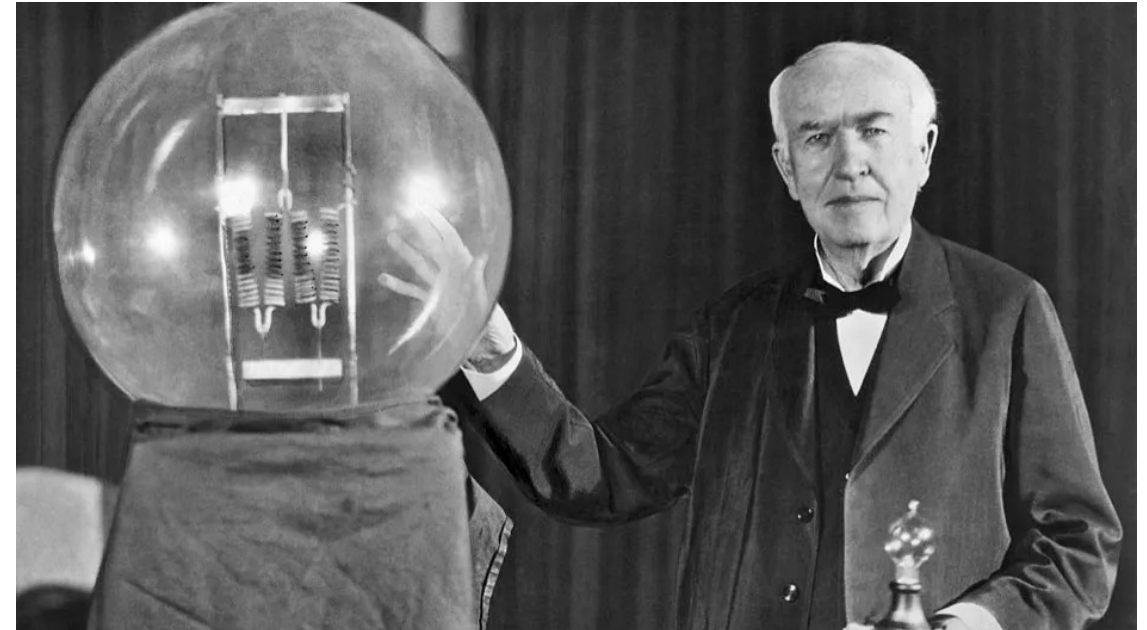
Doing Something Original:

These ideas are novel—based on new observations, unique problems, or creative combinations of disciplines.

Where do research ideas come from?

Doing Something Original:

Thomas Edison: Light Bulb (1879)



Where do research ideas come from?

Doing Something Original:

Thomas Edison: Light Bulb (1879)

Momofuku Ando (Nissin): Instant Ramen (1958)



Where do research ideas come from?

Doing Something Original:

Thomas Edison: Light Bulb (1879)

Momofuku Ando (Nissin): Instant Ramen (1958)

Denso Wave: QR Code (1994)



Where do research ideas come from?



Where do research ideas come from?

Common sense:



Where do research ideas come from?

Common sense:

The ability with which animals (including humans) process sense-perceptions, memories and imagination in order to reach many types of basic judgments.

Where do research ideas come from?

Common sense:

The ability with which animals (including humans) process **sense-perceptions**, **memories** and **imagination** in order to reach many types of basic judgments.

Where do research ideas come from?

Common sense

Why

Your answer

How

Expectations

Collect/Analyse data

What it means

What it doesn't mean

Where do research ideas come from?

Model

Research Question

Develop a theory

Identify variables (if applicable)

Identify hypotheses

Test the hypotheses

Evaluate the results

Critical Review



Common sense

Why

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Critical Review

Observation: Your desktop PC monitor turns black and the PC is silent as well.

Question: What is wrong with the PC?

Hypothesis: Power cut, and hence the PC shutdown
(alternate hypotheses: PC crashed, PC is in hibernate mode)

Prediction: If the power went off, then the fan should not work either.

Test: Turn on the fan to check if it is indeed power cut.

Conclusion: The hypothesis is either proved or disproved (if disapproved, then go to alternate hypothesis and test)

Where do research ideas come from?

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Research Question

Develop a theory

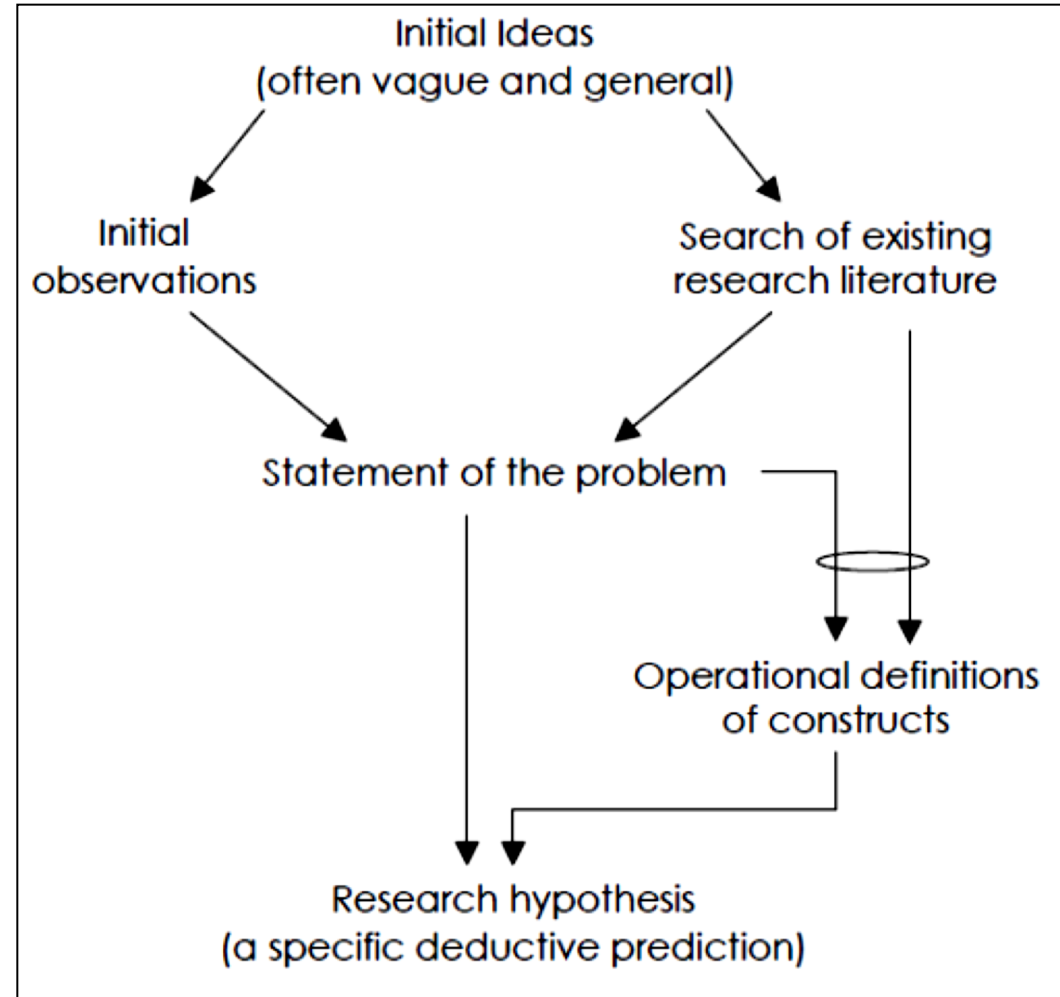
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Where do research ideas come from?

Model

Research Question
Develop a theory
Identify variables (if applicable)
Identify hypotheses
Test the hypotheses
Evaluate the results
Critical Review

Common sense

Why
Your answer
How
Expectations
Collect/Analyse data
What it means
What it doesn't mean

Research Paper

Intro
Intro
Method
Method
Results
Conclusion
Conclusion



Where do research ideas come from?

RPW1: Research Paper Writing!

Why should we write a report, thesis or publish a paper? Why should there be a guidelines for this?

Research Paper

Intro

Intro

Method

Method

Results

Conclusion

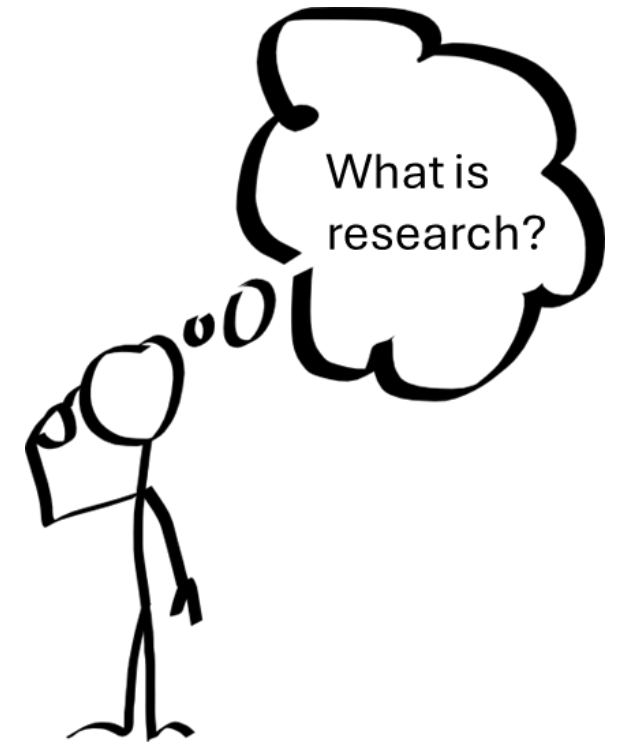
Conclusion

Where do research ideas come from?

Research is: Quantifiable, Verifiable, Replicable and Defensible!

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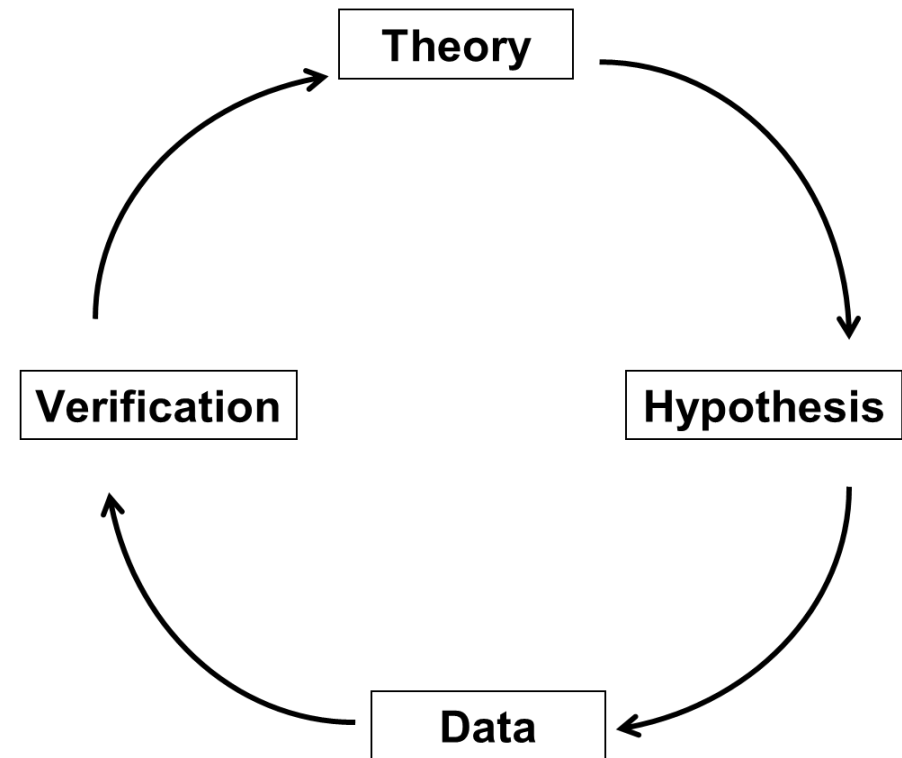
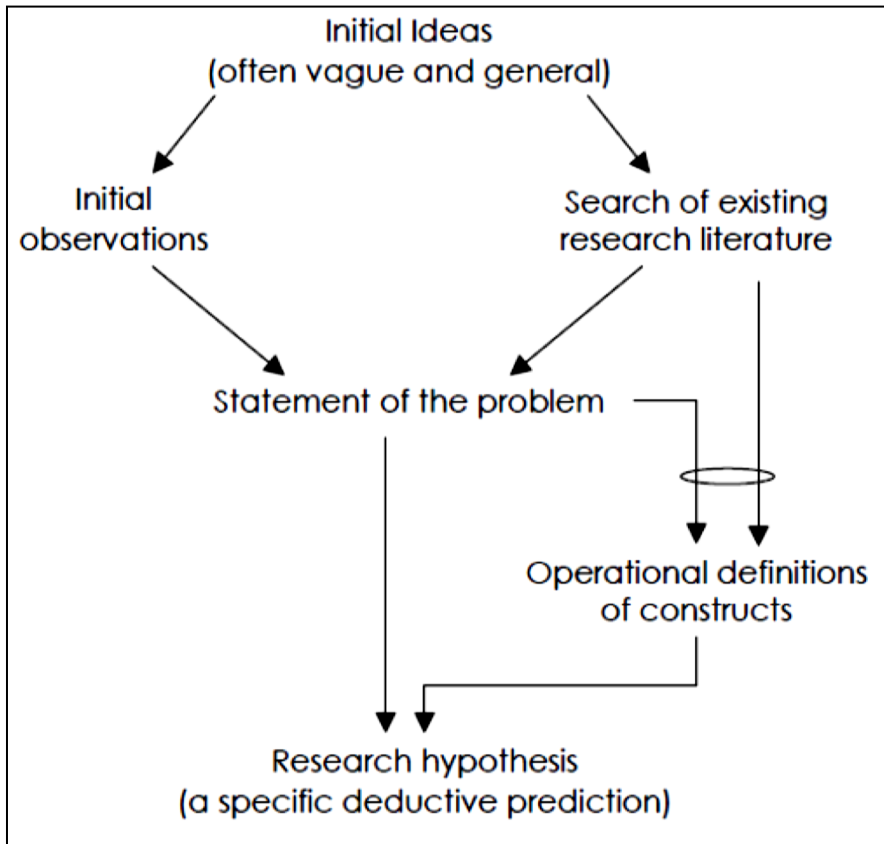


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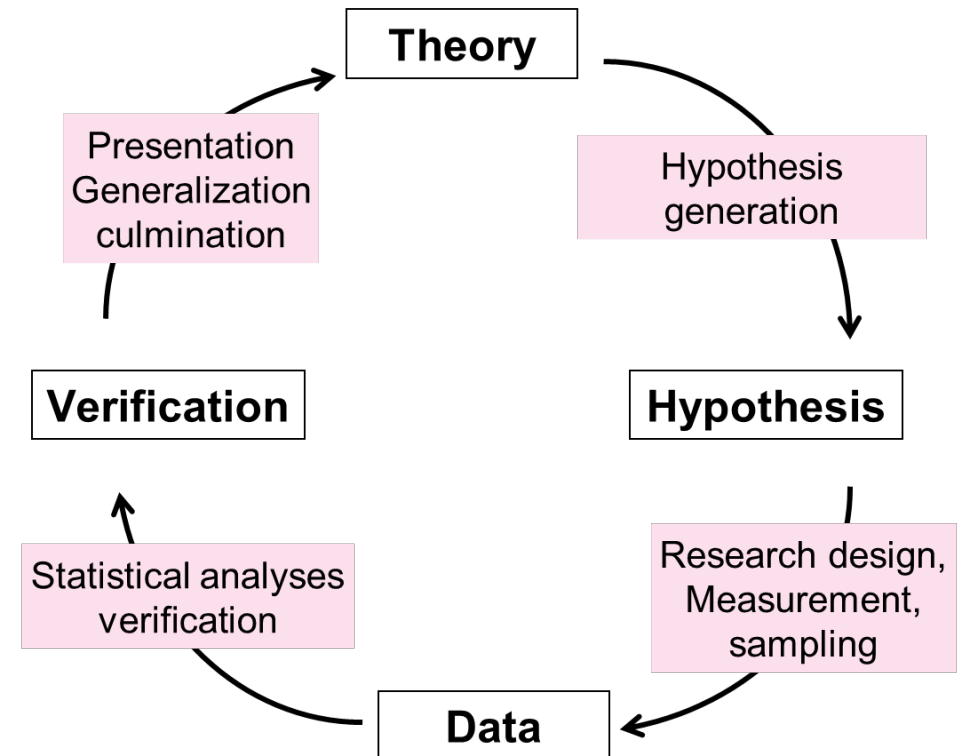
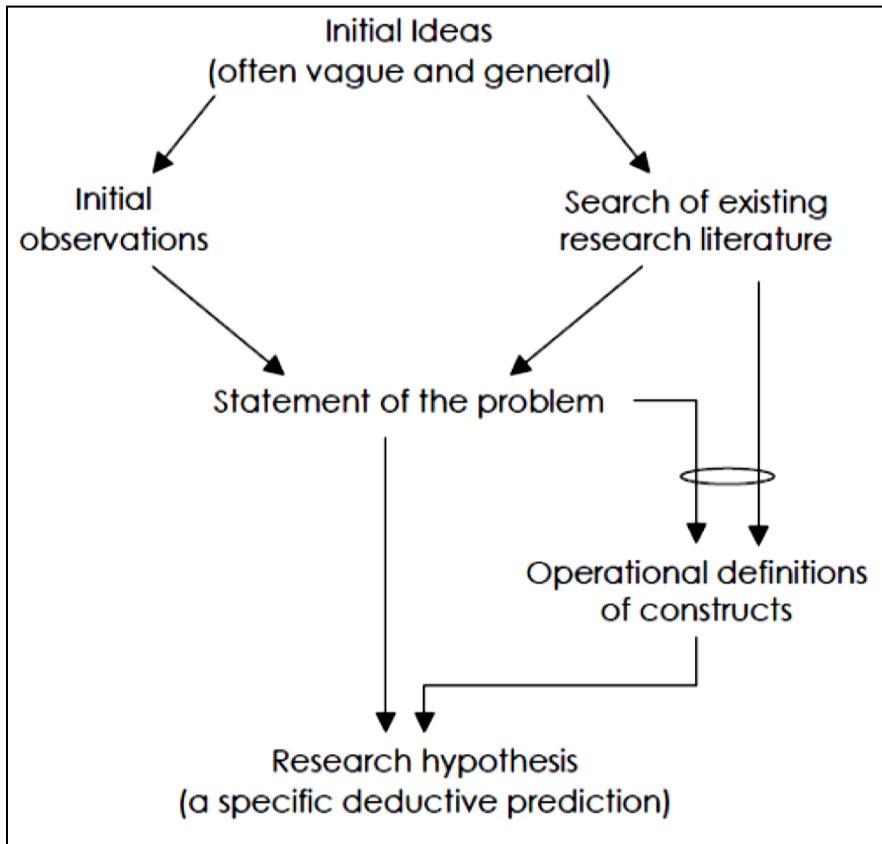
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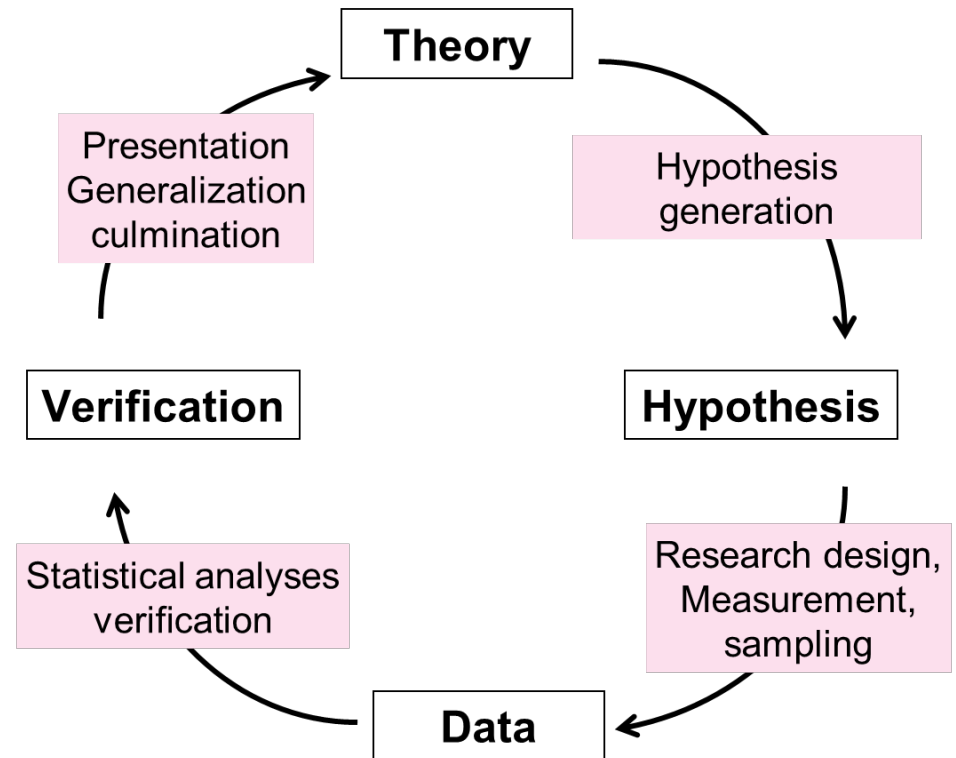
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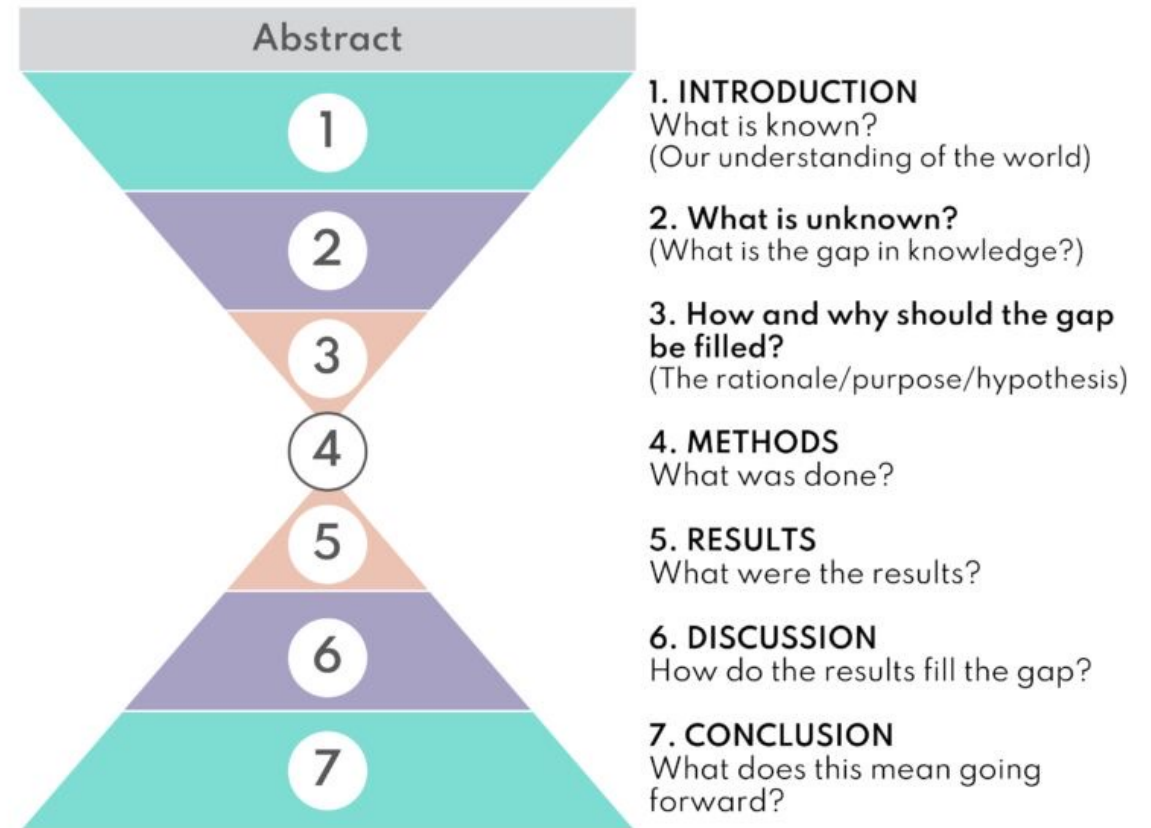
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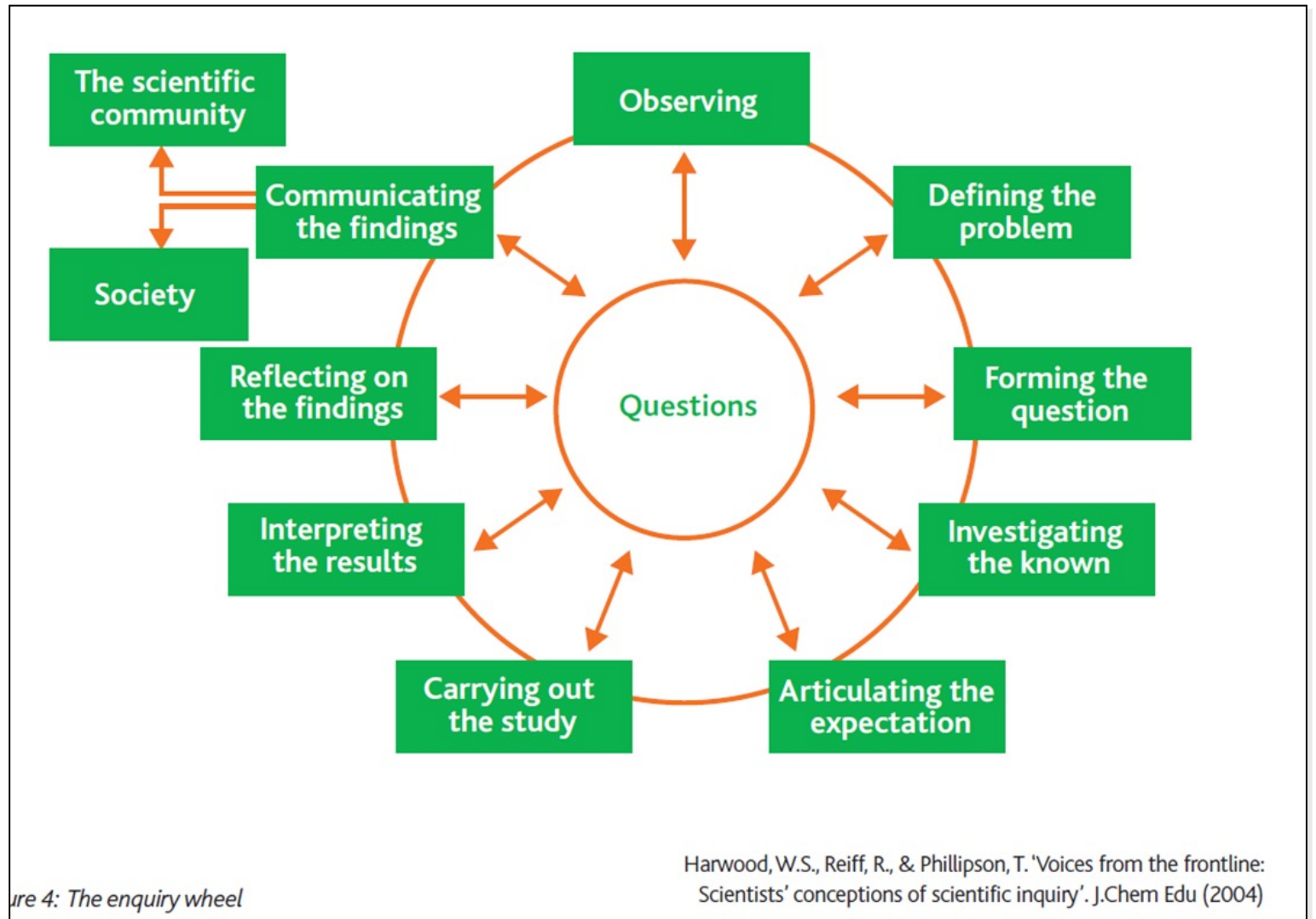
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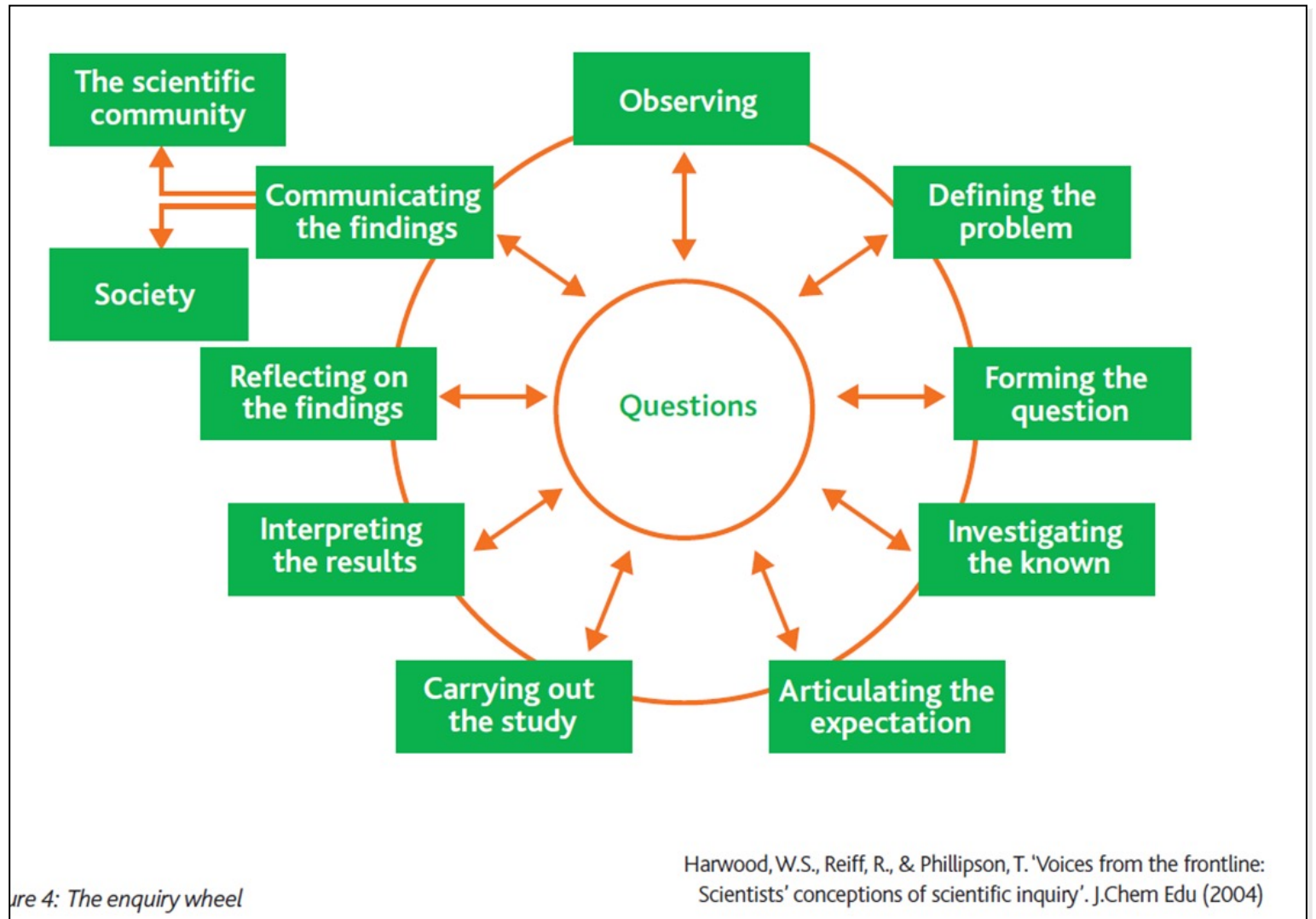
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The Enquiry Wheel!



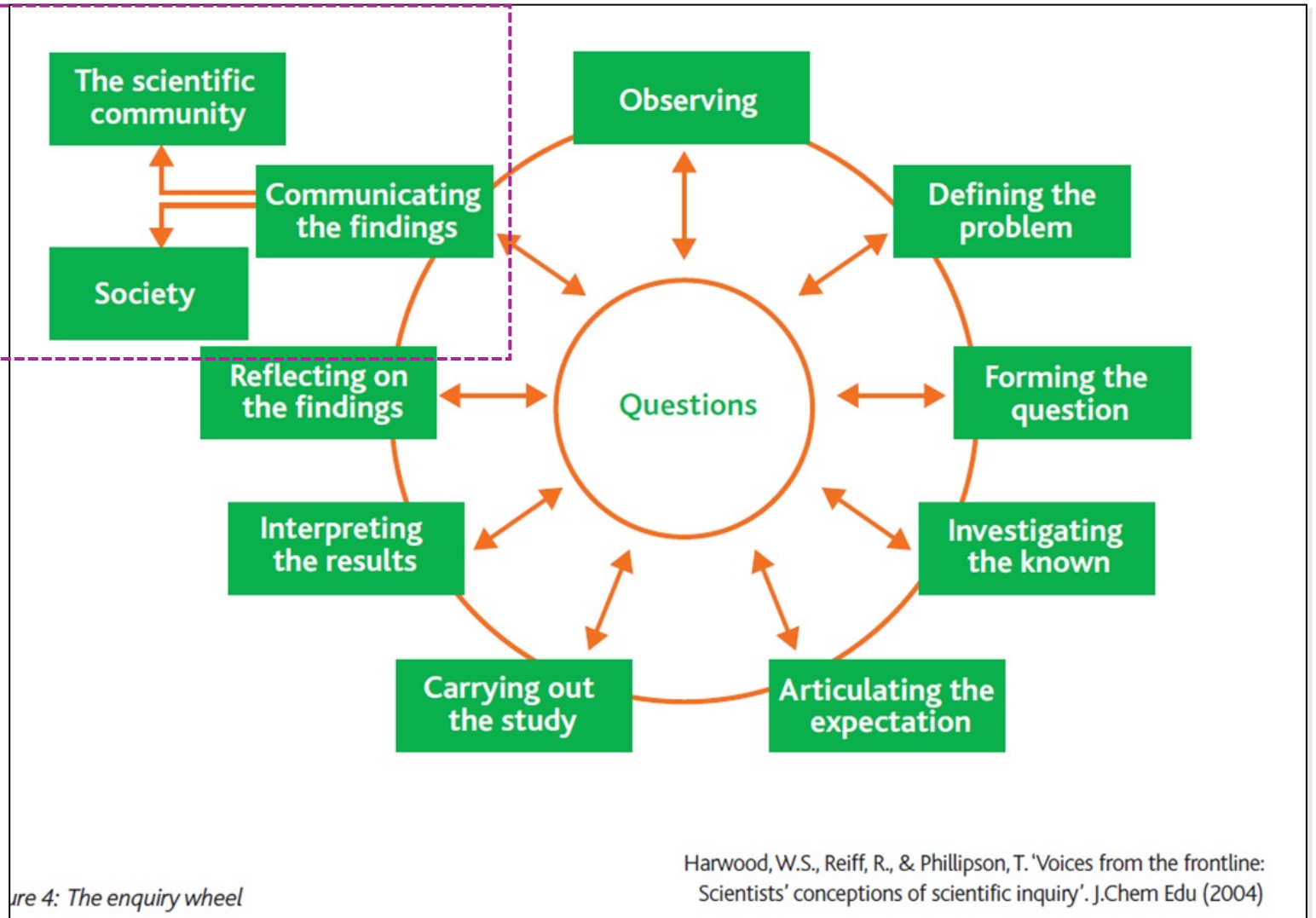
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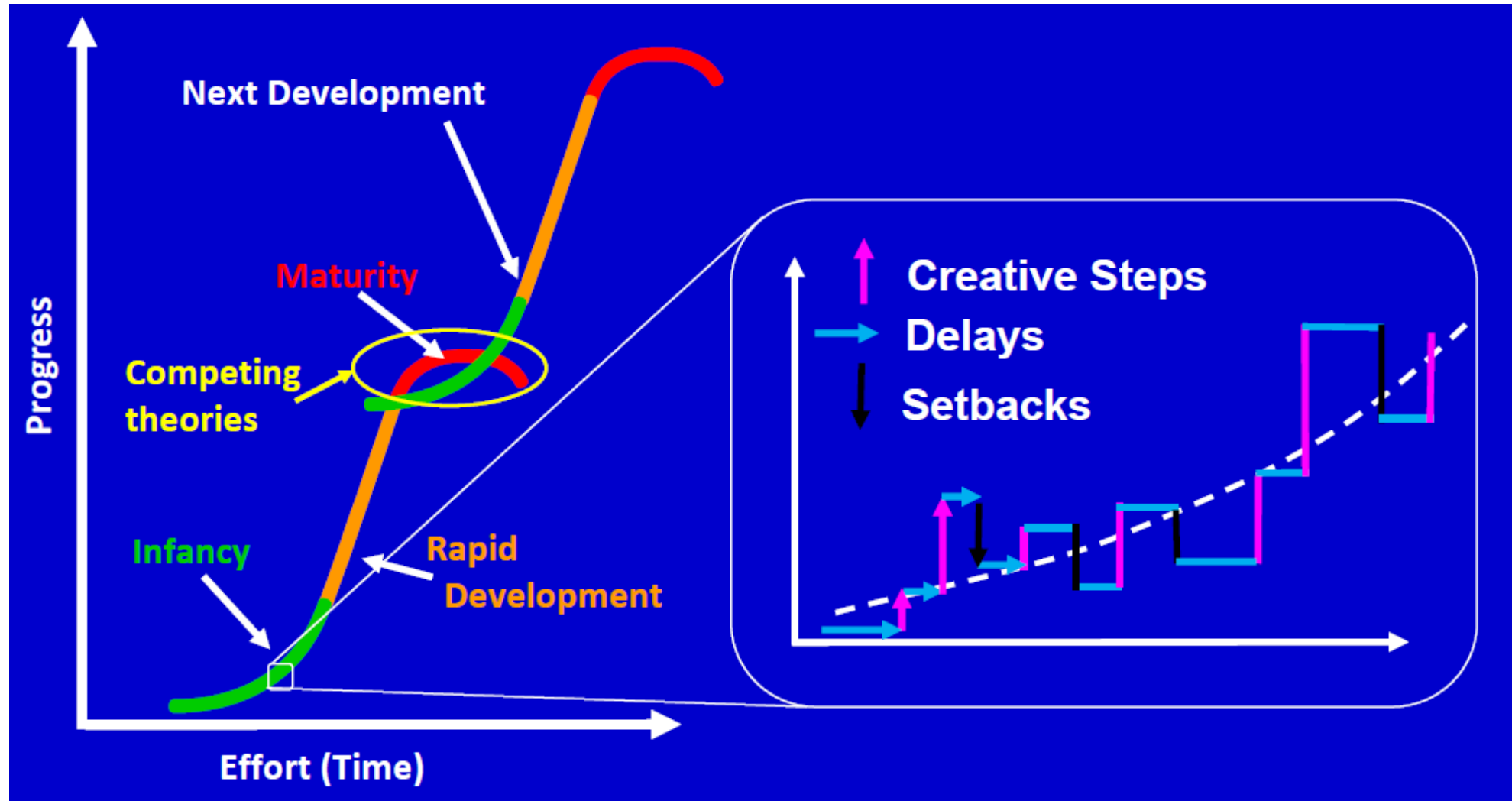


Where do research ideas come from?

RPW1: Research Paper Writing!



Where do research ideas come from?



Where do research ideas come from?

Feature	Adaptive Thinking	Creative Thinking
Definition	Modifying existing ideas to fit new contexts	Generating something new and original
Purpose	Optimization and flexibility	Innovation and disruption
Mindset	"How can I make this work here?"	"What's never been done before?"
Risk Level	Moderate (based on known models)	High (unproven ideas)
Outcome	Improved efficiency and better fit	Breakthroughs and new paradigms
Example	Customizing smartphone features for elderly users	Inventing the first smartphone

Where do research ideas come from?

Adaptive



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Example	Customizing smartphone features for elderly users	Inventing the first smartphone

Creative



Where do research ideas come from?

Adaptive

Where do you stand?

Creative



more adaptive

more creative



Where do research ideas come from?

Adaptive

Where do you stand?

Creative



more adaptive

more creative



Where do research ideas come from?

- Read
- Ask question
- Plan experiments
- Conduct experiments
- Analyze results
- Conclude outcome
- Write the findings
- Present the findings
- Defend the findings



Where do research ideas come from?



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